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India is the second most populous country in the world with 934 million people and according to estimates, it will grow to 1.4 billion by 2030. However population growth is not simply a problem of numbers; it is a problem of human welfare and development. This paper contends that population growth is not the only, or even the primary, source of low levels of living, eroding self-esteem and limited freedom in the less developed nations. Contrary to customary assumptions, population growth in conjunction with other determinants of development has on many instances promoted social change, and in the recent past has been a boon to economic growth in the newly industrialised countries. It is therefore an issue of management and optimum utilisation of present and future human resources.

THE global population will increase during the next 35 years to around 8.5 billion with almost 90 per cent of the increase occurring in developing countries; additionally, about 70 per cent of the increase will occur in poor countries where the average person's income is less than two dollars a day. India, the second most populous country in the world with 934 million people will grow to 1.4 billion by 2030, and China with its 1.2 billion people will continue to remain the population giant in 2030 with 1.5 billion [World Bank 1994].

These increases are unprecedented in the history of mankind (Figure 1), but this phenomenal growth of population is not simply a problem of numbers. It is a problem of human welfare and development. However, the alarmist exhortation against the tidal wave of population increase makes little sense if attempts are not made towards a comprehensive understanding that would integrate demographic and economic as well as other developmental assumptions.

If development entails improvement in people's level of living – their income, education, health and their general well-being – and if it also includes their self-esteem, respect, dignity and freedom to choose, then the important question about population increase is: How does the present population situation in many developing countries contribute to or detract from their chances of realising the goals of development, not only for the current generation but also for future generations? Conversely, how does 'development' affect population growth?. Consequently, the major issues relating to this basic question are:

(i) What are the implications of higher population growth rates among the world's poor for their chances of overcoming the human misery of absolute poverty? Will world food supply and its distribution be sufficient to meet the needs of the anticipated population increase in the coming decades?

(ii) How will the developing countries cope with the vast increases in their labour forces over the coming decades? Will employment opportunities be plentiful, or will it be a major achievement just to prevent unemployment levels from rising?

(iii) To what extent are the low levels of living an important factor in limiting the freedom of parents to choose a desired family size? Is there a relationship between poverty and family size?

(iv) Will the developing countries be capable of improving the levels of living for their people with the current and anticipated level of population growth? To what extent does rapid population increase make it more difficult/easier to provide essential social services including housing, transport, sanitation and security?

(v) To what extent is the growing affluence and the desire to grow further among the economically more developed nations an important factor preventing poor nations from accommodating their growing population? Is the relentless pursuit of increasing affluence among the rich an even more detrimental force to rising living standards among the poor than the absolute increase in their numbers [Todaro 1989]?

With reference to the above questions, it becomes essential to analyse the population issue not simply in terms of numbers, or movement of people, or rates or densities, but as social trends which if favourable and well-managed open a man's options and enlarge his choices.

In the subsequent part of the paper I contend that population growth is not the only, or even the primary, source of low levels of living, eroding self-esteem and limited freedom in the less developed nations. At best, rapid population growth in many countries and regions may appear to be a serious intensifier or multiplier of integral components of underdevelopment. The position espoused in the essay essentially opposes the view expressed in 1968 by Robert McNamara who, on becoming president of the World Bank, called population growth the "greatest single obstacle to economic and social advancement in most of the societies of the developing world". This extreme version of population as a serious problem attempts to attribute almost all of the world's economic and social evils to excessive population growth. It is claimed to be the principal cause of poverty, low levels of living, malnutrition and ill-health, environmental degradation, and a wide variety of other social problems. Incendiary and value-laden words such as the 'population bomb' or 'population explosion' are often used, and dire predictions of food shortages and ecological disaster are attributed almost entirely to the growth in world numbers.

The essay is structured on three general lines of argument which endeavour to establish the idea that rapid population growth in itself is not a serious development problem as often mistakenly perceived by planners and policy analysts.

The Real Problem

(a) *Problem of low levels of development is not population* per se *but other determinants of social change:*

The main thrust of this premise is, if strategies that lead to improved standard of living are promoted and pursued, population will cease to appear as detrimental. Thus, underdevelopment is the real problem and 'development' should be the goal.

Consequently, to achieve higher levels of living it is important to raise the overall levels of economic growth – not only for private consumption but also for improved public services.

The links between population and economic growth, especially food productivity has for long been the subject of intense controversy, but writers such as Julian Simon and Ester Boserup have asserted that population growth is beneficial, or even essential, to long-term economic growth. To illustrate this line of thinking , in Europe before 1800,[1] the living standards of the majority usually fell in periods of population growth, but on the other hand large-scale population declines were also observed during periods of improved living standards. This phenomenon suggests that the productive limits of cultivable land had largely been reached; that population growth could be combined with improved living standards only if technology - to produce more food, housing, goods and services from the same amount of land – advanced: and that technology did not, as a rule, move fast enough to keep pace.

The United Nations Food and Agriculture Organisation (1985), had observed that between 1974 and 1983 per capita food production in developing countries increased almost twice as rapidly as in developed countries. This occurrence refuted the earlier unduly pessimistic position taken by the UN when it questioned the ability of the less developed countries to produce sufficient food to keep pace with population growth.

Besides affecting the availability of land and the amount of labour expended on it, population size and growth also influence its productivity. Although the expansion of agricultural land in the developing countries has been a significant factor in the growth of agricultural output, higher yield of crop output was the predominant factor. In this regard, Boserup's study (1965) indicated that population growth and the pressure it exerts are important stimuli for increasing agricultural yields. Further, a more recent study by Boserup (1981) as stated by Horlacher and Mackellar (1988), points out that population increase, which expands food requirements, also tends to produce increased food supplies by bringing about a shift towards more intensive land use. The 'green revolution' in Pakistan and in the north-western India are remarkable examples of this striking increase in agricultural output to meet the challenge of a rapidly growing population (Figure 2) by way of introduction of improved high-yielding variety of seeds and superior methods of land use.

Pakistan's 'green revolution' contributed to other aspects of economic development as well. First, the new technology not only produced a much higher output per man, but it also led to an increase in employment in the agricultural sector. The increase in employment took place because more land could be cultivated and high-yielding seeds required more labour in preparing the seed beds, applying fertilisers and pesticides, weeding, harvesting and threshing. More labour was also needed because new techniques permitted multiple cropping as has been observed in the Indian state of Uttar Pradesh [Elkan 1973 and Agarwal 1985:91].

Mellor and Johnston (1984) have projected that the rate of growth of food production will continue to exceed the rate of population growth. This belief was largely based on the assumption that in many developing countries the infrastructural and other necessary requirements for accelerated agricultural growth are now in place, while population growth rate is likely to decline in future. However, in many developing countries the task of meeting the nutritional requirements of a growing population will require an increase in the population supporting capacity of their agricultural land. With regard to this, Wilkinson (1973) argues: "Development is primarily the result of attempts to increase the output from the environment rather than produce a given output more efficiently." Therefore population size – the quantitative dimension of biological man – and its impact decide the amount of labour which is available and the quantity of food and other materials needed; it can almost be regarded as the most important determinant of the scale of cultural operations which relates man to the production systems.

Additionally, apart from having a quantitative impact on the agricultural social system, the size of the population also has a qualitative impact. The limitation on the supply of resources can mean that quantitative change necessitates qualitative change. The 'carrying capacity' is a function of the kind of productive technology which is used. Subsequently, to increase the carrying/supporting capacity of any territory a new adaptive strategy based on new technology will need to be introduced, as has been noted in the 'green revolution' in India and Pakistan. Thus, on the whole it can be stated that much of the historical development of agricultural techniques the world over can be interpreted as a series of attempts to increase the carrying capacity of available land area in response to population growth [Wilkinson 1973].

Desirable Phenomenon

(b) *For many developing countries and regions population growth has been and is a desirable phenomenon:*

Recent economic-demographic simulation models tend to emphasise the role of population growth as an incentive for capital formation as compared to earlier models which laid stress on the negative impact of population growth on the capital-labour ratio. Nowadays, it is believed that by making labour more plentiful, population growth can raise the return on capital relative to the return on the real wage rate, thus encouraging capital formation. The increased rate of return on capital resulting from population growth might also be expected to encourage direct foreign investment. This becomes obvious from the table showing the seven Asian countries presently enjoying the world's highest economic growth rates.

Studying the total fertility rate (TFR) of these countries 25-30 years ago, which was relatively quite high, and the simultaneous decline in the infant mortality rate (IMR) in these years, there must have been a substantial population increase in the young population of these countries, now resulting in some 70 per cent of the seven's population being under 30 years. Moreover, as the TFR has declined in the last 30 years, the child dependency ratio[2] has also decreased, leaving a large section of the population in the labour force who are highly productive as well as voracious consumers, leading to capital formation and setting the economy towards a boom. The fact that population growth rates change slowly and their effect on economic growth may take many years to become apparent, is also revealed in the table. The relationship between population growth and per capita income growth involves both current and lagged components.

Table: Demographic Details and Economic Performance of Seven Asian Countries

	Countries	H K	IND	MAL	S K	SIN	TAI	THA	INDIA
PGR	25-30 yrs	2.6	2.2	2.9	2.5	2.4	–	3.1	2.3
PGR	15.20 yrs	2.2	2.4	2.3	1.9	1.5	–	2.7	2.3
PGR	MRE	0.7	1.8	2.9	1.0	2.9	–	0.6	2.1
CBR	25-30 yrs	27.3	42.7	40.4	35	30.7	–	41.2	44.8
CBR	15-20 yrs	18.1	38.4	32.1	25.9	17.7	–	32.5	37.5
CBR	MRE	12.6	25.7	30.3	16.1	17.3	–	22.2	30
TFR	25-30 yrs	4.5	5.5	6.3	4.9	4.7	–	6.3	6.2
TFR	15-20 yrs	2.5	5.01	4.6	3.3	2.1	–	4.6	5.3
TFR	MRE	1.5	3.1	3.8	1.8	2	–	2.5	3.9
CDR	25-30 yrs	5.7	20.2	11.6	11.3	5.5	–	10	20.3
CDR	15-20 yrs	4.9	15.4	8.5	7.4	5.1	–	8.3	15.1
CDR	MRE	5.5	8.5	5.3	6.2	5.2	–	7	10.6
IMR	25-30 yrs	27	127.6	55.2	62.2	26.1	–	88.4	150
IMR	15-20 yrs	14.9	108.6	37.2	39.8	13.9	–	55	130
IMR	MRE	6.6	61.2	15.9	17.1	6.6	–	27.2	92
GNP	25-30 yrs	620	30	320	130	530		140	90
GNP	15-20 yrs	2180	210	820	580	2550	–	360	170
GNP	MRE	11490	570	2320	5400	11160	–	1420	350
	A (Per Cent)	5.4	6.4	8.5	7.5	6.0	6.8	8.6	–
	B (Per Cent)	18	14.3	18	11.4	9.5	9.2	12.3	

Key: MRE: Most Recent Estimate; PGR: Population Growth Rate, in Annual Percentage; CBR: Crude Birth Rate, per thousand population. TFR: Total Fertility Rate in births per woman; CDR: Crude Death Rate, per thousand population; IMR: Infant Mortality Rate, in thousand live births; GNP: Per Capita: Gross National Product (MRE = 1990), in US $; A: Growth Projected 1992 GNP increase; B: Export Projected 1992 Growth.

Countries: H K: Hong Kong; IND: Indonesia; Mal: Malaysia; S K: South Korea; SIN: Singapore; TAI: Taiwan; THA: Thailand.

Sources: *Social Indicators of Development 1991-92* (published for the World Bank; John Hopkins University Press, Baltimore) and *Time Magazine*, September 14, 1992 (from Pacific Economic Co-operation Council, Population Reference Bureau)

The past population growth is reflected in an increase in the current labour force, whereas comparatively a smaller rate of current population growth, consisting primarily of additional children, will add to the denominator of per capita income.

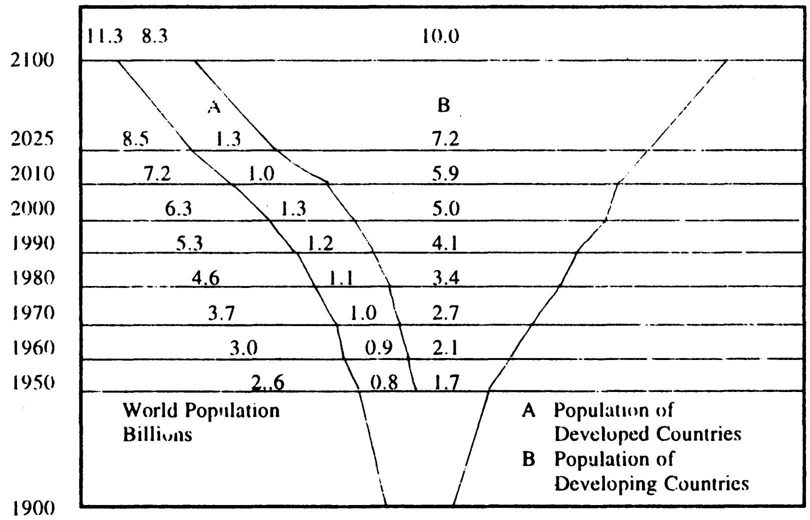
Furthermore, there is usually a concern that the capital stock will not expand rapidly enough to absorb into the modern sector the increase in the labour force resulting from rapid population growth. In most developing countries, however, those who do not find employment in the modern sector will find work in the other sectors rather than join the ranks of the unemployed. The informal sector promotes skills that are a prerequisite for advancing into modern sector employment, apart from being productive [Sinclair 1978; Horlacher and Mackellar 1988].

The price mechanisms by which population growth affects economic development are highly complex. However, it can be reasoned that one of the routes is through their effects on savings and investment. In poor countries, having additional children is a form of saving in itself: parents forego present income in order to provide for their old age just as people in developed countries put money aside for pensions. But as the need for extra children declines as an old age security, funds that were to be spent on children now become available as savings for other type of investment, thus setting the ground for economic growth. This is another illustration of population and income growth which involves both current and lagged components. It can also be speculated that countries with a large young population such as India and other developing nations, a shift from a high- to a low-childbearing pattern is usually one in which there are more young households saving for retirement relative to the number of elderly households drawing down accumulated assets [Mason 1988].

Interestingly, a study by the National Academy of Sciences in 1986 pointed out that some investments in land improvement may require a certain minimum population density before they can become profitable. The same has been argued by some third world neo-Marxist pro-natalists, that many rural regions in developing countries are in reality underpopulated, in the sense that much unused but arable land could yield large increases in agricultural output if only more people were available to cultivate it. Many regions of tropical Africa and Latin America and even parts of Asia are said to be in this situation. With respect to Africa, for example, some observers have noted that many regions had larger populations in the remote past than what exists today. Rural depopulation resulted not only from the slave trade but also from compulsory military service, confinement to reservations and the forced labour policies of former colonial governments. For example, in the 16th

FIGURE 1: WORLD POPULATION GROWTH BY REGIONS

Growth of World Population

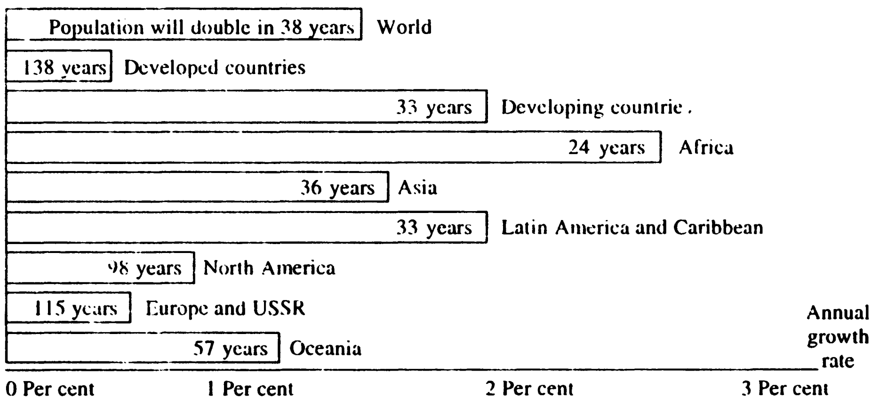


Figures are based on the latest projections. These are rather higher than those projected until recently, due to slowing down in the decline of fertility rates in a number of countries.

Population by region, in billions

Region	1990	2025	2100	Region	1990	2025	2100
Africa	0.8	1.6	3.0	North America	0.3	0.3	0.3
Asia	3.1	4.9	8.3	Europe and USSR	0.8	0.9	0.8
Latin America and Caribbean	0.5	0.8	0.9	Oceania	0.03	0.04	0.04

Annual growth rate as measured in 1990 and time for the population to double



Source: Population, Resources and the Environment, United Nations Population Fund, 1991.

century the Congo kingdom is said to have had a population of approximately two million. But by the time of the colonial conquest, which followed 300 years of slave trade, the population of the region had fallen to less than one-third of that figure. Today's Zaire has barely caught up to the 16th century figure. Other regions of western and eastern Africa provide similar examples [Todaro 1989]. Comparably, Pingali and Binswanger (1987) attribute the failure of some large-scale irrigation schemes in sub-Saharan Africa to the sparseness of the population and a corresponding lack of demand.

With reference to Clark, Simon and Eberstadt [Todaro 1989, chapter 6], a conventional argument is that population growth is an essential ingredient to stimulate economic development. Larger populations provide the needed consumer demand to generate favourable economies of scale in production, to lower production costs, and to provide a sufficient and low cost labour supply to achieve higher output levels and larger markets, which should permit greater specialisation and division of labour, reduced cost per head of public goods of various kinds and reduced costs per head of transportation, communication networks, and other forms of social structure.

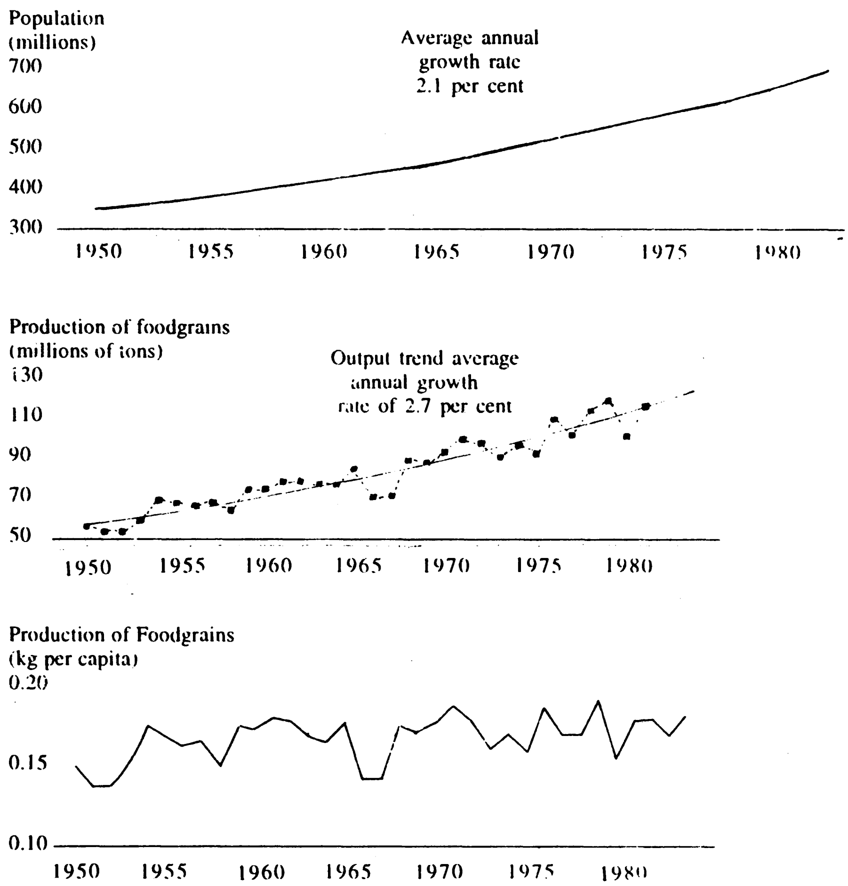
Although the possibilities of beneficial effects of population growth through economics of scale is generally acknowledged. there is less agreement as to where these economies are likely to be found. For instance, in the US, rising population densities stimulated the development of the transport system during the 19th century. An improved transport system in turn greatly facilitated the growth of agriculture by lowering transport costs and by raising the farm-gate prices of agricultural products, whereas the potential benefits to be gained from higher population densities was not realised in rural Bangladesh: transport, marketing and storage facilities, as well as extension services, were all inadequate [World Bank 1984]. In this context, the National Academy (1986:52), asserts that "manufacturing economies of scale exist principally at the urban level and are exhausted at a moderate level of city size", while "economies of scale are likely to occur in agriculture, especially by spreading the fixed cost of infrastructure and research over a large number of people".

Insofar as population growth increases demand, it can contribute to the realisation of economies of scale. This argument is found to be true in the case of the British Virgin Islands. In these islands from 1980-91, the population grew at an annual rate of about 4 per cent and if the growth rate is allowed to continue unchecked, the population would double every 17.3 years. This increase in population affected the economy positively as there was a considerable increase in working age group primarily due to in-migration, due to which remarkable economic progress has been realised. GNP - at factor cost in constant prices - has grown from 77.07 (US $ million) in 1984 to 112.44 (US $ million) in 1989, an increase of about 9.2 per cent per annum, and per capita income has increased from about 5,800 (US $) in 1984 to about 7,130 (US $) in 1989, an increase of about 4.6 per cent per annum [British Virgin Islands, Statistical Abstract No 3, 1990]. Hence, it can be inferred that population growth in conjunction with other socio-economic determinants provided the necessary stimulus for favourable economies of scale to occur in these islands.

Therefore, it makes sense when population 'rivisionist', neoclassical economists of the 1980s such as Julian Simon and Nicholas Eberstadt argue, for instance, that a free market will always adjust to any exigency created by population. For example, scarcities will drive up prices and signal the need for new cost-saving production technologies. Thus, free market and human ingenuity will solve any and all problems arising from population growth. This revisionist perspective is clearly in contrast with the traditional 'orthodox' argument of the 1950s to 1970s that rapid population growth had serious economic consequences and if left unchecked would slow down economic development.

FIGURE 2

Foodgrains and Population in India, 1950-83



Years shown are end years for the agricultural season. Data for 1983-84 are estimates
Source: World Bank data adapted from Cassen 1976

Source: The World Development Report, 1984.

Population and Natural Resources

(c) Myth of population pressure and depletion of natural resources:

The environmental dimension to population is firmly grounded in economics, among other factors. Behind the demographic issues of population growth and its uneven distribution, fertility levels, age dependency ratios, migration patterns and urbanisation, lies the pressing concern of economic advancement and sustainable development. Therefore, the search for sustainability necessitates that the population issue be addressed within the context of natural resource management (Figure 3).

The concept of 'carrying capacity' has often been used as an effort to build a bridge between the environment and population issues. To quote Kingsley Davis (1991): "The general idea is that by analysis of soil and other elements of the environment, one can estimate how many people can be supported on a given amount of land". Also, a definition by T N Srinivasan, "the maximum population that can be sustained indefinitely into the future", cited by Davis, provides a good basis for contesting this point.

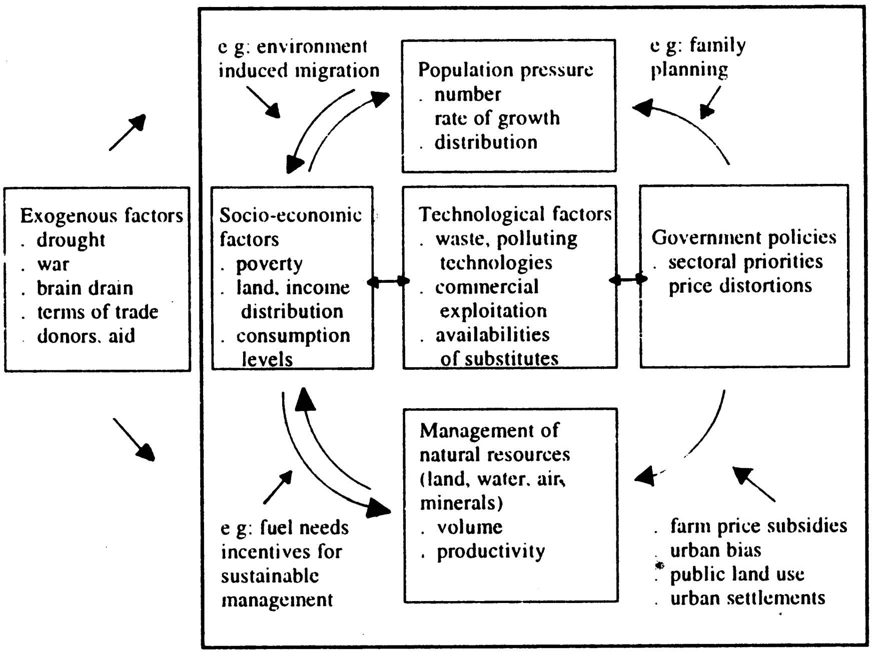
First, why should the definition be preoccupied only with the 'maximum' population? Why not consider the 'minimum' population as well? For instance, in the counties of western Texas, US, there are not enough people to man the existing infrastructure. The same individual may therefore have to occupy more than one post. Yet few would venture to say that this population is too small to be 'sustained'. An animal species runs the danger of suffering extinction if its number falls below a critical level, but hardly anyone would reckon today that this could possibly be true of human species as well in certain circumstances. As a matter of fact, population in itself is rarely the problem. Its scarcity or density in a region in relation to resources make people adjust their institutions and behaviour to the situation as elucidated subseqently in two unlike conditions.

The immediate effect of the increased resource demand resulting from population growth would be higher resource prices. In these circumstances, it would encourage the exploration of ecologically friendly sources of energy and stimulate research on alternative technology. There is evidence that the development of renewable resources, such as commercial forests, had responded to price signals in a similar situation. Further, increasing natural resource prices and improving technology might lead to adaptation in institutional arrangements, for example, the conversion of common property resources to private property resources thereby promoting conservation and improved management [Horlacher and Mackellar 1988]. With the help of this argument can be defended the remarkable success of village forest protection commitees (FPCs) in West Bengal, India. In 1972, the forest department recognised its failures in reviving degraded sal forests in the south western districts of the state. The traditional methods of surveillance and policing had led to a complete alienation of the people from the administration. The department changed its strategy, making a beginning in the Arabari forest range of Mednipur district. Here, the villagers were involved in the protection of 1,272 ha of badly degraded sal forests. In return for their help in protection, they were employed in both silvicultural and harvesting operations, given 25 per cent of the final harvest and allowed fuel, food and fodder collection on payment of a nominal fee. With the active participation of the villagers, the sal forests showed a remarkable recovery, and in barely 10 years, a previously valueless patch was estimated to be worth Rs 12.5 crore [Gadgil and Guha 1992:26-29].

Additionally, consumers would be persuaded to adjust to less resource-intensive goods as natural resource prices increase and producers adopt less resource-intensive methods of production. Therefore, there are reasons to believe that population pressure on scarce natural resources, usually resulting from population density and growth, can be an important decisive factor associated with natural resource development and management of an area. Unlike the above example, Australians and Canadians do not find their sparse populations to be an obstacle to maintaining a high level of living. Thus, the presence or absence of poulation is not the obvious problem.

Second, the 'carrying capacity' supposedly refers to the commodities wholly or partially necessary for the individuals to live, but from the context in which it is usually referred to one concludes that the commodity really being considered is food. For the very same logic, Davis (1991) has called the usual 'carrying capacity' resoning the 'the bread alone fallacy'. The writers on population issues concentrate on food not only because it is necessary for existence, but it is intimately, continuously, and appetitively involved with the individual organism. It is therefore viewed as a proxy for population itself. Thus, to analyse why a given region is populous, an observer may simply point out that the soil there is singularly fertile. Consequently, the special role of food in an individual's life makes it easy, in reasoning about population and resources, to confuse a necessary cause (food) with a sufficient one.

FIGURE 3: LINKAGES BETWEEN DEMOGRAPHIC AND NATURAL RESOURCE ISSUES



Source: Population, Resources and the Environment, United Nations Population Fund, 1991.

The in-built food bias in 'carrying capacity' leads to other distortions as well, for example, in unwarranted and largely unconscious assumptions regarding the nature of demographic changes. The biggest error of this kind is the tendency to automatically consider mortality as the chief mechanism by which human numbers are adjusted to resources. As an illustration, if the discussion concerns contemporary sub-Saharan Africa, the usual reasoning takes the following sequence: If the production of food in central Africa continues to fall, the result will be more famines, which will drive up the death rate and thus slow down the rate of population increase. Little account is taken of population being regulated by other important factors such as migration or fertility.

The fixation on food — the assumption that self-sufficiency in food production is desirable — gives rise to another misinterpretation. Contrast this trend of thinking with actuality, where for good reason international trade in foodstuffs has been increasing. Since people must eat every day and since most foodstuffs are perishable or seasonal, international trade in food may have more advantage than it does in many other commodities. As a result, if a country has certain advantages in non-food trade, it can use these to buy food. Therefore, a policy of self-sufficiency in food makes no sense for countries like Taiwan, Japan, the US or any other country that has access to international trade. The earth is to be perceived as one large trading system and so the 'carrying capacity' must refer not just to food that could be grown within the national borders, but to the total of what could be gained in all ways [Davis 1991].

Further, many times the idea of 'carrying capacity' is treated as a norm or optimum that should be sought. For instance, to admit that the world can support 33 billion[3] people and not more is to imply that below that figure there would be enough to eat for everyone, but above the mark of 33 billion people would starve. Clearly, the underlying assumption is, as the increase in population reaches a high level, it automatically will stop it self – primarily through diseases caused mainly by malnutrition. Hence, according to this idea, people are perceived as automations passively waiting to be killed by their own reproductive success. Overlooked is the possibility that the increase in human numbers may be stopped by people themselves at a point far below the theoretical 'carrying capacity' not essentially because of a scarcity of food and other requirements, but for the love of the frills and foibles of advanced civilisation and a distaste for the arduous tasks of bearing and rearing children.

CONCLUSION

It can be asserted that although in the 1980s income in countries whose population had grown more slowly in the 1965-80 period grew 1.8 per cent faster than in countries with faster 1965-80 population growth – unlike most studies of developing countries in which no connection could be found between the growth rates of population and of per capita incomes up to around 1975 – the statistical link is probably an expression of several factors [United Nations Population Fund 1992].

Though the strength of the population factor can be judged by comparing it with the most obvious alternative explanation for the slow income growth in the 1980s, the high debt levels, no correlation could be found at all in the 82 countries between the debt levels in 1980 and the growth in incomes over the 1980s [UNPF 1992]. Consequently the explanation to the statistical link has to be explored in other determinants. One is the way in which rapid population growth commandeers a large share of investment merely to maintain the same amount of capital per person, leaving little left over to improve capital per person. However, more widely the link almost certainly expresses the impact of other factors that contribute both to slower population growth and faster economic growth, especially higher levels of education and health in the workforce, and increased access of women to the labour market.

To conclude, for any particular level of population growth there is always a big gap between the best and worst economic pêrformance. Good economic and trade policies can combat adverse circumstances, whereas bad ones can fail even in favourable situations. Even if the recent figures on economic and population growth do suggest, at the very least, that rapid population growth since 1965 may have created problems for economic development in the more difficult climate of the 1980s, good management can diminish the population effect to a considerable extent and has the potential of transforming this impediment to growth into a boon. Conversely, a review of studies in this field affirms, where management was poor, rapid population growth has magnified other difficulties. Thus, the question to be addressed seriously is that of good or poor management of population and not of population *per se*.

Notes

1 The famous English monk Thomas Malthus, states: (1) population is necessarily limited by the means of subsistence, and (2) population invariably increases where the means of subsistence increase, unless prevented by some very powerful and obvious checks.

2 One frequently used index summarising an age distribution is known as the dependency ratio. Strictly, this is the ratio of economically active to economically inactive persons in the population, but often, because of lack of data or difficulties in defining economic activity in many countries, a ratio of age group is used instead. Then the ratio is:

$$\text{Dependency Ratio} = \frac{\text{Children+ Elderly}}{\text{Working Ages}} \times 100$$

Therefore, ratios using only children, or only the elderly in the numerator, are also sometimes calculated. Accordingly, it is referred to as 'child dependency ratio' or 'old age dependency ratio'.

3 The estimated 33 billion human beings under stipulated conditions was the result of a major study of this type by FAO, UNFPA and IIASA. The following quotations describe the procedure adopted: The study "combines a climate map.. with soil map... and then divides the study area into grids of 100 sq kms each.. the 15 most widely grown food crops were considered... Three alternative levels of farm technology were postulated." Given these estimates of potential production, along with the calorie and protein requirements of the human body as recommended by FAO and WHO, the investigators computed "the maximum population which can be supported" [Srinivasan 1988: 14-15, cited in Davis 1992]

The findings were as follows:

Farm Technology	Billions of People: Carrying Capacity	Billions of People: Projected in 2000
Low	5.6	3.6
Intermediate	14.9	3.6
High	33.2	3.6

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